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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

6433349721

PHYSICS **5054/33**

Paper 3 Practical Test **October/November 2010**

2 hours

Candidates answer on the Question Paper.

Additional Materials: As listed in the Confidential Instructions.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions.
For each of the questions in Section A, you will be allowed to work with the apparatus for a maximum of 20 minutes. For the question in Section B, you will be allowed to work with the apparatus for a maximum of 1 hour.
You are expected to record all your observations as soon as these observations are made.
An account of the method of carrying out the experiments is **not** required.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
Total	

This document consists of **11** printed pages and **1** blank page.

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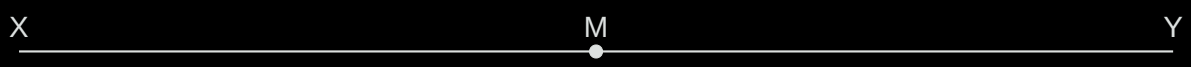


Fig. 2.1

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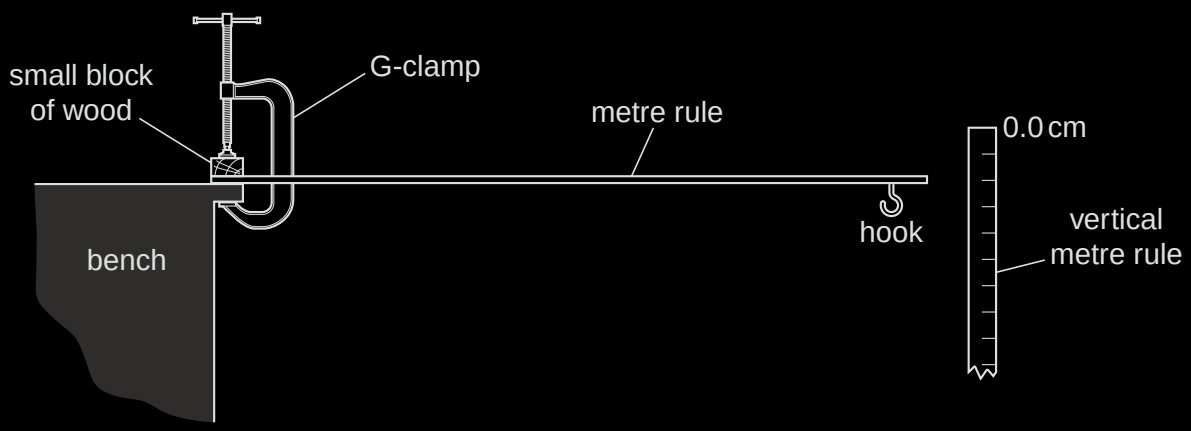


Fig. 4.1

Explain, with the aid of a diagram, how you ensured that the metre rule was vertical.

.....

.....

..... [1]

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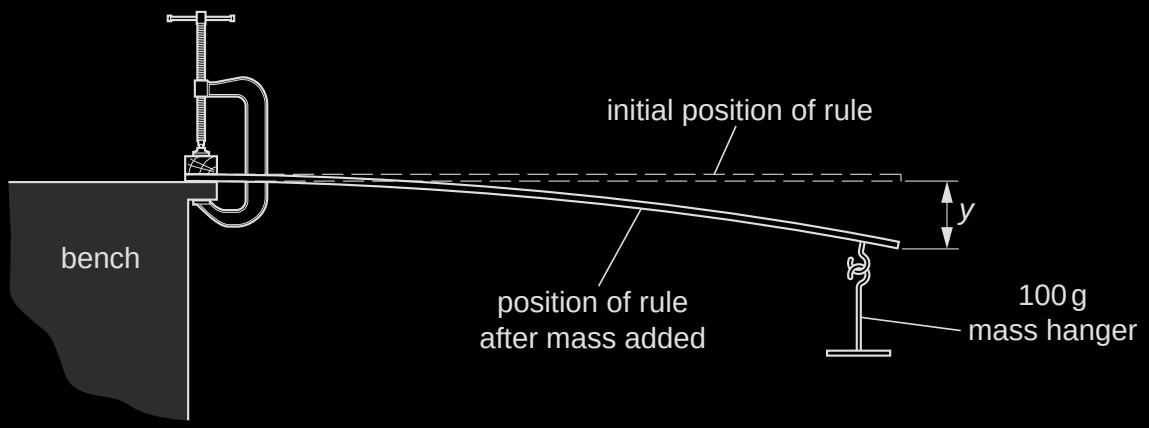


Fig. 4.2

(ii) Determine the vertical displacement *y* of the end of the horizontal rule, as shown in Fig. 4.2. Show the two readings that you took from the vertical rule in order to determine *y*.

y = ... [2]

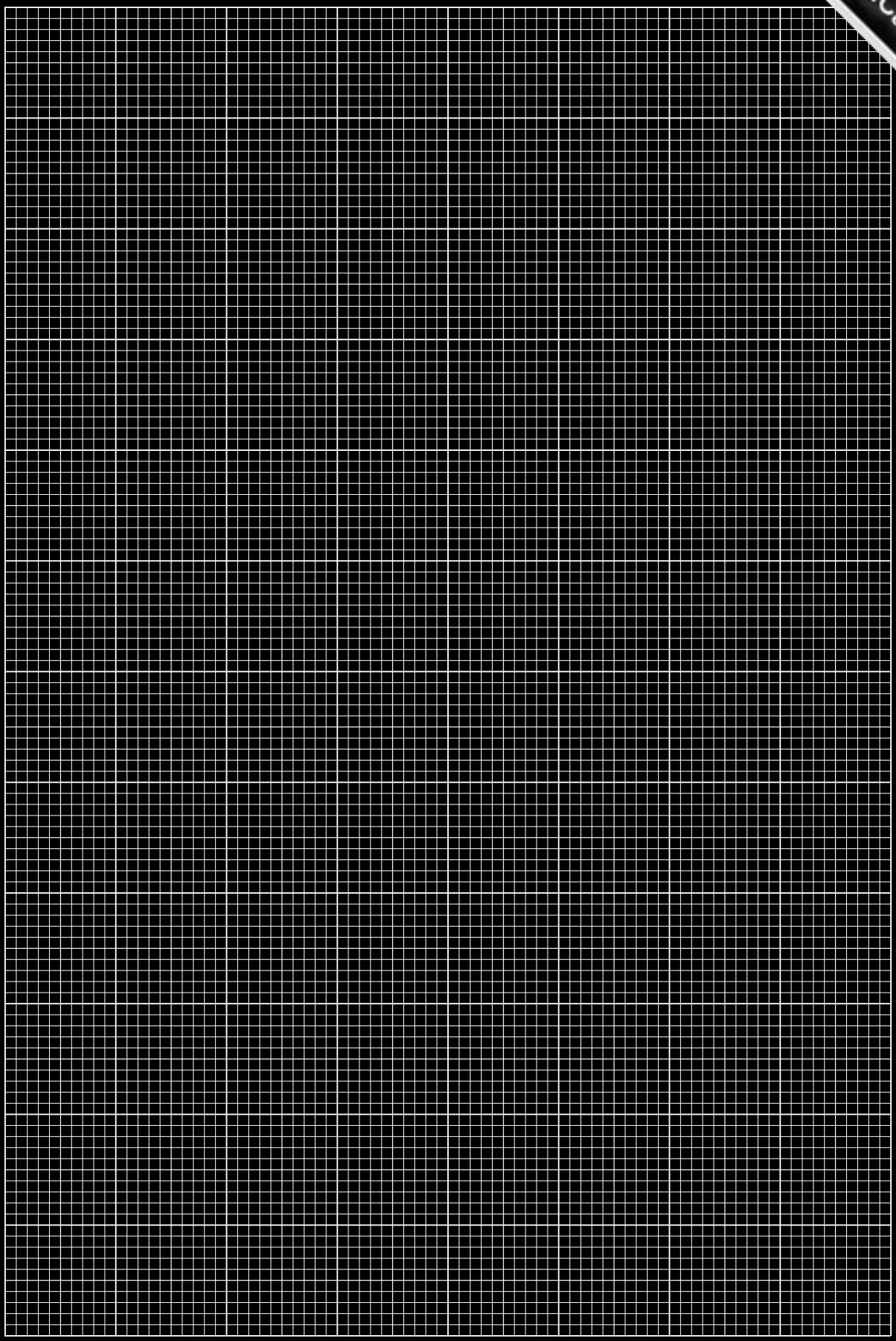
(c) Add a further 100 g to the mass hanger and measure the total vertical displacement of the end of the horizontal rule from its starting position when no load was applied.
Record the total mass *M* suspended from the rule and the new value of *y*. Show the readings that you took in order to determine the total vertical displacement.

M = ...

y = ...
[2]

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